

Power MOSFET

Complementary H-Bridge Switching Pair

Part Numbers: IRFZ44N (N-Channel) & IRF9530N (P-Channel)

This report documents both MOSFETs identified directly inside the uploaded Proteus design file (ROOT.DSN): an **IRFZ44N** N-Channel HEXFET Power MOSFET and an **IRF9530N** P-Channel HEXFET Power MOSFET. The schematic explicitly references the component descriptions "IRFZ44N" / "irfz44n.pdf" and "IRF9530N" / "irf9530n.pdf" as linked datasheet files, confirming exact part identification (no assumption was required for these two components). The two devices form a complementary switching pair used together in the H-Bridge DC Motor Driver with PWM circuit.

1. Component Name

Designator (Proteus)	Part Number	Polarity	Package
Q (High-side switches)	IRF9530N	P-Channel Enhancement HEXFET	TO-220AB
Q (Low-side switches)	IRFZ44N	N-Channel Enhancement HEXFET	TO-220AB

2. Introduction

A Power MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor) is a voltage-controlled switching device used to control large currents and voltages in power electronic circuits using a very small gate-drive signal. The IRFZ44N and IRF9530N are HEXFET-technology power MOSFETs originally developed by International Rectifier (now part of Infineon Technologies). The IRFZ44N is an N-Channel device that conducts when a positive gate-to-source voltage is applied, while the IRF9530N is its P-Channel counterpart, which conducts when the gate is driven negative with respect to the source. Using one of each type allows a push-pull (H-Bridge) switching arrangement to be built without needing a dedicated high-side driver for every leg of the bridge.

3. Manufacturer

Parameter	IRFZ44N	IRF9530N
Manufacturer	International Rectifier (Infineon Technologies)	International Rectifier (Infineon Technologies)
Official Datasheet	IRFZ44N / IRFZ44NPbF HEXFET Power MOSFET	IRF9530N HEXFET Power MOSFET (PD-91482)
Datasheet Ref.	www.infineon.com/assets/row/public/documents/24/49/infineon-irfz44n-datasheet-en.pdf	PD-91482C, irf.com/IRF9530N

4. Functional Description

Working Principle

Both devices operate as enhancement-mode MOSFETs. Applying a gate-source voltage above the threshold voltage ($V_{GS(th)}$) creates a conductive channel between drain and source, allowing current to flow with a very low on-resistance ($R_{DS(on)}$). Removing the gate voltage collapses the channel and turns the device off. Because switching is controlled purely by gate voltage (not continuous current, as with a BJT), the MOSFET requires negligible steady-state drive power and can switch at high frequency, which is essential for

PWM-based motor speed control.

Purpose in this Circuit

The IRFZ44N/IRF9530N pair act as electronically-controlled power switches that connect the DC motor terminals to the positive supply rail and ground in the correct combination to drive the motor forward, reverse, or brake, while a PWM signal rapidly switches one arm of the bridge on and off to regulate the average voltage (and therefore speed) delivered to the motor.

Why used in this H-Bridge DC Motor Driver with PWM project

Discrete power MOSFETs were chosen (rather than an integrated driver IC such as an L298N) because they can handle higher continuous current with much lower conduction losses ($R_{DS(on)}$ in the tens of milliohms), giving higher efficiency and less heat generation for the same motor current. Their fast switching characteristics are also well suited to the PWM frequency generated by the STM32F103C6 microcontroller in this design.

5. Key Features

Feature	IRFZ44N (N-Channel)	IRF9530N (P-Channel)
Drain-Source Voltage (VDSS)	55 V	-100 V
Continuous Drain Current (ID)	49 A @ 25 degC	-14 A @ 25 degC
Static On-Resistance RDS(on)	17.5 m-ohm (max, VGS=10V)	0.20 ohm (max, VGS=-10V)
Max. Power Dissipation	94 W (approx.)	88 W
Gate Threshold Voltage	2.0 V to 4.0 V	-2.0 V to -4.0 V
Operating Temperature	-55 degC to +175 degC	-55 degC to +175 degC
Package Type	TO-220AB	TO-220AB
Switching	Fast switching, dynamic dv/dt rated	Fast switching, dynamic dv/dt rated
Protection	Fully avalanche rated, integral reverse body diode	Fully avalanche rated, integral reverse body diode

6. Pin Configuration (TO-220AB, both devices)

Pin Number	Pin Name	Function
1	Gate (G)	Control input; voltage applied here (relative to Source) turns the channel on/off
2	Drain (D)	Main current-carrying terminal (also bonded to the metal tab)
3	Source (S)	Reference terminal; current return path / connection to load or ground

7. Electrical Specifications

IRFZ44N (N-Channel) — Electrical Characteristics @ Tj = 25 degC

Parameter	Symbol	Value	Conditions
Drain-Source Breakdown Voltage	V(BR)DSS	55 V (min)	VGS = 0V, ID = 250uA
Static Drain-Source On-Resistance	RDS(on)	17.5 m-ohm (max)	VGS = 10V, ID = 25A
Gate Threshold Voltage	VGS(th)	2.0-4.0 V	VDS = VGS, ID = 250uA

Parameter	Symbol	Value	Conditions
Continuous Drain Current	ID	49 A	TC = 25 degC
Total Gate Charge	Qg	63 nC (max)	ID = 25A
Drain-Source Leakage Current	IDSS	250 uA (max)	VDS = 44V, TJ = 150 degC
Thermal Resistance Junction-Ambient	RthJA	62 degC/W (max)	-

IRF9530N (P-Channel) — Electrical Characteristics @ Tj = 25 degC

Parameter	Symbol	Value	Conditions
Drain-Source Breakdown Voltage	V(BR)DSS	-100 V (min)	VGS = 0V, ID = -250uA
Static Drain-Source On-Resistance	RDS(on)	0.20 ohm (max)	VGS = -10V, ID = -8.4A
Gate Threshold Voltage	VGS(th)	-2.0 to -4.0 V	VDS = VGS, ID = -250uA
Continuous Drain Current	ID	-14 A	TC = 25 degC
Diode Forward Voltage (body diode)	VSD	-1.6 V (max)	TJ = 25 degC, IS = -8.4A
Reverse Recovery Time	trr	130-190 ns	TJ = 25 degC, IF = -8.4A

8. Typical Applications

- H-Bridge / half-bridge DC motor drivers
- Pulse Width Modulation (PWM) based speed controllers
- DC-DC converters and switch-mode power supplies (SMPS)
- Battery-powered load switching circuits
- Automotive 12 V electronic control circuits
- Relay-replacement solid-state switching
- LED lighting drivers and dimmers
- Inverter and power-conversion front-ends

9. Advantages

- Very low on-resistance minimizes conduction power loss
- Voltage-controlled gate requires negligible steady-state drive current
- Fast switching enables efficient high-frequency PWM operation
- Rugged, fully avalanche-rated for inductive load switching (e.g. motors)
- Wide operating temperature range suitable for varied environments
- Low cost and wide commercial availability (industry-standard TO-220 package)

10. Limitations

- Gate is sensitive to electrostatic discharge (ESD) and requires careful handling
- P-Channel devices (IRF9530N) have notably higher RDS(on) than equivalent N-Channel parts, increasing conduction losses
- Requires adequate heat-sinking at higher continuous currents
- Gate voltage must be sufficiently high (near 10 V) to fully saturate the channel; direct 3.3 V microcontroller drive is not sufficient without an intermediate gate-drive stage

- Body diode reverse-recovery can generate switching losses in hard-switched bridge topologies

11. Role in This Project

In the H-Bridge DC Motor Driver with PWM project, the four MOSFETs (two IRFZ44N low-side devices and two IRF9530N high-side devices) form the four arms of the H-Bridge that surround the DC motor. The STM32F103C6 microcontroller generates PWM control signals that are applied to the gates of the selected diagonal pair of MOSFETs, switching them on and off rapidly. By modulating the duty cycle of this PWM signal, the average voltage delivered to the motor — and hence its rotational speed — is controlled, while the choice of which diagonal pair is switched determines the direction of rotation.

12. Precautions

- Handle using standard ESD precautions (anti-static wrist strap/mat) as the gate oxide can be damaged by static discharge
- Never exceed the maximum gate-source voltage rating (typically +/-20 V)
- Provide adequate heat-sinking when the motor draws current close to the continuous drain current rating
- Include flyback/free-wheeling protection across the motor terminals to protect the MOSFETs from inductive voltage spikes
- Ensure the gate is never left floating; use a pull-down/pull-up resistor to hold the device off during power-up
- Verify PCB clearances for the higher voltage rating of the IRF9530N (-100 V) during layout

13. Official Reference

Item	IRFZ44N	IRF9530N
Official Manufacturer	Infineon Technologies (formerly International Rectifier)	Infineon Technologies (formerly International Rectifier)
Official Datasheet Title	IRFZ44NPbF / IRFZ44N HEXFET Power MOSFET	IRF9530N HEXFET Power MOSFET
Document / Version	Document No. 3363173 / PD-9.1737 (rev. varies by source)	PD-91482C